

Tick-Borne Infections – Summary

Key Symptoms:

- **Early disease:** Rash (can be bullseye shaped but not always), low-grade fever, fatigue
- **Early disseminated (<3 months):** Muscle aches, fatigue, encephalitis/meningitis, cardiac conduction issues
- **Late disease (>3 months):** Confusion, fatigue, encephalopathy, multiple joint arthritis, etc.

Most common causing-pathogens:

- **Lyme disease:** *B. burgdorferi*

Diagnosis: Clinical symptoms/history, PCR and/or antibody serology screening

Prevention:

- Wear closed-toe shoes, light coloured long-sleeve shirts & pants. Pull socks over pants.
- DEET or Icaridin-containing insect repellents
- Tick check → rapid removal if found, save tick for testing (only useful if local Public Health is actively testing ticks)
- Shower or bathe within 2 hours of being outdoors to wash away loose ticks
- When removing a tick DO NOT twist the head or jerk (may detach head from body), grab as close to your skin as possible and pull slowly out

Post-Exposure Prophylaxis :

- Single dose doxycycline
- Eligibility: tick attached at least 24-36 hours, tick removed within last 72 hours, bite acquired in highly endemic area, *Ixodes* spp. tick (commonly referred to as blacklegged ticks), no contraindication to doxycycline
- Note: A single dose of doxycycline is considered safe for children of all ages and for pregnant women

Treatment:

- Early disease (Erythema Migrans): Doxycycline **OR** Cefuroxime **OR** Amoxicillin x 10-14 days
 - 2nd line: azithromycin x 7 days
- Late/disseminated Lyme disease: Doxycycline **OR** Cefuroxime **OR** Amoxicillin **OR** Ceftriaxone x 14-28 days
 - Adjunctive treatments: NSAIDs, acetaminophen, OT, psychology/psychiatry

Additional Key Points:

- First generation cephalosporins are not effective in the treatment of Lyme disease
- Symptom resolution may take several months, especially in late disease